34.6: Mammals are amniotes that have hair and produce milk

An evolutionary tree highlights mammalia.

Branch point 1: First level-1 branch: Cephalochordata. The second level-1 branch splits from branch point 2 into the first level-2 branch: Urochordata. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Myxini and the second level-4 branch: Petromyzontida. The second level-3 branch splits from branch point 5 into the first level-5 branch: Chondrichthyes. The second level-5 branch splits from branch point 6 into the first level-6 branch: Actinopterygii. The second level-6 branch splits from branch point 7 into the first level-7 branch: Actinista. The second level-7 branch splits from branch point 8 into the first level-8 branch: Dipnoi. The second level-8 branch splits from branch point 9 into the first level-9 branch: Amphibia. The second level-9 branch splits from branch point 10 into the first level-10 branch: Reptilia and the second level-10 branch: Mammalia.

The reptiles we have been discussing represent one of the two living lineages of amniotes. The other amniote lineage is our own, the **mammals**. Today, there are about 6,400 known species of mammals on Earth.

Derived Characters of Mammals

Mammals are named for their distinctive mammary glands, which produce milk for offspring. All mammalian mothers nourish their young with milk, a balanced diet rich in fats, sugars, proteins, minerals, and vitamins. Hair, another mammalian character, and a fat layer under the skin provide insulation that can conserve water and protect the body against extremes of heat or cold. Another mammalian adaptation for life on land is the kidney (see Figure 44.13), which is efficient at conserving water when removing wastes from the body. Some mammals, such as kangaroo rats, are so adept at conserving water that they can survive in arid environments while drinking little or no water at all **(Figure 34.37)**.

Figure 34.37

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    Adaptations of the kangaroo rat to its extremely dry
habitat.
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Kangaroo rat in its underground burrow with descriptions of five adaptations that enable it to survive.

Five key adaptations that enable the kangaroo rat to survive in its dry habitat.

1.The kangaroo rat's thick, oily skin limits evaporative water loss.

2.The animal stays in its cool, relatively humid burrow during the heat of the day, emerging at night to forage.

3.Due to the unusual shape of its nasal passages, very little water is lost when the animal exhales.

4.A kangaroo rat does not need to drink: It obtains all of the water it needs from catabolic pathways and from moisture in food.

5.The large intestine and kidney absorb water so effectively that little water is lost in feces and urine.

MAKE CONNECTIONS *Explain how the catabolic pathways mentioned in* **4** *could provide a kangaroo rat with water. (See*Concept 9.1*.)**

*For suggested answer, see*Appendix A*.**

Like birds, mammals are endothermic, and most have a high metabolic rate. Efficient respiratory and circulatory systems (including a four-chambered heart) support a mammal's metabolism. Also as in birds, mammals generally have a larger brain than other vertebrates of equivalent size, and many species are capable learners. The relatively long duration of parental care extends the time for offspring to learn important survival skills by observing their parents. In addition, whereas the teeth of reptiles are generally uniform in size and shape, the jaws of mammals typically bear a variety of teeth with sizes and shapes adapted for chewing many kinds of foods. Humans, like most mammals, have teeth modified for shearing (incisors and canine teeth) and for crushing and grinding (premolars and molars).

Early Evolution of Mammals

Mammals belong to a group of amniotes known as **synapsids**. Early nonmammalian synapsids lacked hair, had a sprawling gait, and laid eggs. A distinctive characteristic of synapsids is the single temporal fenestra, a hole behind the eye socket on each side of the skull. Humans retain this feature; your jaw muscles pass through the temporal fenestra and anchor on your temple. Fossil evidence shows that the jaw was remodeled as mammalian features arose gradually in successive lineages of earlier synapsids (see Figure 25.7); in all, these changes took more than 100 million years. In addition, two of the bones that formerly made up the jaw joint (the quadrate and the articular) were incorporated into the mammalian middle ear **(Figure 34.38)**. This evolutionary change is reflected in changes that occur during development. For example, as a

mammalian embryo grows, the posterior region of its jaw—which in a reptile forms the articular bone—can be observed to detach from the jaw and migrate to the ear, where it forms the malleus.

Figure 34.38

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The evolution of the mammalian ear bones.
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The evolution of the mammalian ear bones.

Biarmosuchus was a synapsid, a lineage that eventually gave rise to the mammals. Bones that transmit sound in the ear of mammals arose from the modification of bones in the jaw of nonmammalian synapsids.

Parts are as follows.

Part a. Skull of Biarmosuchus, an extinct synapsid. Skull has is long and relatively flat on top, with prominent canines and sharp teeth with a thin squamosal bone around the temporal fenestra behind the eye socket, and the meeting of the small quadrate (upper) and articulate (lower) bones forming the jaw joint. The thick dentary bone lines most of the lower jaw but does not meet the articular bone at the jaw joint.

Part b. During the evolutionary remodeling of the mammalian skull, a new jaw joint formed between the dentary and squamosal bones (see Figure 25.7). No longer used in the jaw, the quadrate and articular bones became incorporated into the middle ear as two of the three bones that transmit sound from the eardrum to the inner ear. In the present-day reptile, sound waves can enter the ear and encounter the eardrum. Behind the eardrum is the middle ear, with the stapes bone perpendicular to the eardrum. Beyond the middle ear is the inner ear. In the present-day mammal, sound waves enter the ear and encounter the eardrum. Beyond the eardrum is the middle ear, with the larger malleus (articular) bone connected to the slightly smaller incus (quadrate) bone and then the shortened stapes, forming a structure perpendicular to the eardrum. Beyond the stapes is the inner ear.

MAKE CONNECTIONS *Review the definition of exaptation in*Concept 25.6*. Summarize the process by which exaptation occurs and explain how the incorporation of the articular and quadrate bones into the mammalian inner ear is an example.*

*For suggested answer, see*Appendix A*.*

Synapsids evolved into large herbivores and carnivores during the Permian period (299–252 million years ago), and for a time they were the dominant tetrapods. However, the Permian-Triassic extinctions took a heavy toll on them, and their diversity fell during the Triassic (252–201 million years ago). Increasingly mammal-like synapsids emerged by the end of the Triassic. While not true mammals, these synapsids had

acquired a number of the derived characters that distinguish mammals from other amniotes. They were small and probably hairy, and they likely fed on insects at night. Their bones show that they grew faster than other synapsids, suggesting that they probably had a relatively high metabolic rate; however, they still laid eggs.

During the Jurassic (201–145 million years ago), the first true mammals arose and diversified into many short-lived lineages. A diverse set of mammal species coexisted with dinosaurs in the Jurassic and Cretaceous periods, but most of these species were small, measuring less than 1 m in length. One factor that may have contributed to their small size is that dinosaurs already occupied many ecological niches of large-bodied animals.

Fossil and molecular evidence indicate that by 160 million years ago, the three major lineages of mammals had emerged: those leading to monotremes (egg-laying mammals), marsupials (mammals with a pouch), and eutherians (placental mammals). After the extinction of large dinosaurs, pterosaurs, and marine reptiles during the late Cretaceous period, mammals underwent an adaptive radiation, giving rise to large predators and herbivores as well as flying and aquatic species.

Monotremes

Monotremes are found only in Australia and New Guinea and are represented by one species of platypus and four species of echidnas (spiny anteaters; **Figure 34.39**). Monotremes lay eggs, a character that is ancestral for amniotes and retained in most reptiles. Like all mammals, monotremes have hair and produce milk, but they lack nipples. Milk is secreted by glands on the belly of the mother. After hatching, the baby sucks the milk from the mother's fur.

Figure 34.39

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    Short-beaked echidna (*Tachyglossus aculeatus*), an
Australian monotreme.
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Photograph of a short-beaked echidna covered in bushy spine-like structures and a long, slender snout. Egg is about the size of a person's thumbnail.

Monotremes have hair and produce milk, but they lack nipples. Monotremes are the only mammals that lay eggs (inset).

Marsupials

Opossums, kangaroos, and koalas are examples of the group called **marsupials**. Both marsupials and eutherians share derived characters not found among monotremes. They have higher metabolic rates and nipples that provide milk, and they give birth to live young. The embryo begins development inside the uterus of the female's reproductive tract. The lining of the uterus and the extraembryonic membranes that arise from the embryo form a **placenta**, a structure in which nutrients diffuse into the embryo from the mother's blood.

A marsupial is born very early in its development and completes its embryonic development while nursing **(Figure 34.40a)**. In most species, the nursing young are held within a maternal pouch called a *marsupium*. A red kangaroo, for instance, is about the size of a honeybee at its birth, just 33 days after fertilization. Its back legs are merely buds, but its front legs are strong enough for it to crawl from the exit of its mother's reproductive tract to a pouch that opens to the front of her body, a journey that lasts a few minutes. In other species, the marsupium opens to the rear of the mother's body; in greater bilbies, this protects the young as their mother burrows in the dirt **(Figure 34.40b)**.

Figure 34.40

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Marsupials in Australia and North America.
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Marsupials in Australia and North America.

Parts are as follows.

Part a. A young brushtail possum. The offspring of marsupials are born very early in their development. They finish their growth while nursing from a nipple (in their mother's pouch in most species).

Part b. A greater bilby. The greater bilby is a digger and burrower that eats termites and other insects, along with the seeds, roots, and bulbs of various plants. The female's rear-opening pouch helps protect the young from dirt as the mother digs. Other marsupials, such as kangaroos, have a pouch that opens to the front.

Part c. Virginia opossum (Central and North America). This species is the only marsupial found north of Mexico. After the young complete development in the mother's pouch, they spend another four to five months riding on her back before living independently.

Marsupials existed worldwide during the Mesozoic era, but today they are found only in the Australian region and in the Americas **(Figure 34.40c)**. The biogeography of marsupials illustrates the interplay between biological and geologic evolution (see Concept 25.4). After the breakup of the supercontinent Pangaea, South America and Australia became island continents, and their marsupials diversified in isolation from the eutherians that began an adaptive radiation on the northern continents. Australia has not been in contact with another continent since early in the Cenozoic era, 66 million years ago. In Australia,

convergent evolution has resulted in a diversity of marsupials that resemble eutherians in similar ecological roles in other parts of the world **(Figure 34.41)**. In contrast, although South America had a diverse marsupial fauna throughout the Paleogene, it has experienced several immigrations of eutherians. One of the most important occurred about 3 million years ago, when North and South America joined at the Panamanian isthmus and extensive two-way traffic of animals took place over the land bridge. Today, only three families of marsupials live outside the Australian region, and the only marsupials found in the wild in North America are a few species of opossum.

Figure 34.41

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    Convergent evolution of marsupials and eutherians (placental
mammals).
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Convergent evolution of marsupials and eutherians (placental mammals).

(Drawings not to scale.)

Information is given as a table:

Marsupial Mammals	Eutherian Mammals
Plantigale (rodent like animal with long tail)	Deer mouse (mouse with tail)
Marsupial mole (small animal with unusual facial structure)	Mole (small animal with unusual facial structure)
Sugar glider (resembles a squirrel with long, bushy tail)	Flying squirrel (long, bushy tail, extensions for gliding)
Wombat (mid-sized animal with pointed ears)	Woodchuck (mid-sized animal with rounded ears)
Tasmanian devil (mid-sized animal with sharp teeth and pointed face)	Wolverine (mid-sized animal with pointed face)
Kangaroo (Large animal with long tail, upright stance and pouch)	Patagonian cavy (Large animal with upright stance)

Eutherians (Placental Mammals)

Eutherians are commonly called placental mammals because their placentas are more complex than those of marsupials. Eutherians have a longer pregnancy than marsupials. Young eutherians complete their embryonic development within the uterus, joined to their mother by the placenta. The eutherian placenta provides an intimate and long-lasting association between the mother and her developing young.

The major groups of living eutherians are thought to have diverged from one another in a burst of evolutionary change. The timing of this burst is uncertain: Molecular data suggest it occurred about 100 million years ago, while morphological data suggest it was about 60 million years ago. **Figure 34.42** explores several major eutherian orders and their phylogenetic relationships with each other as well as with the monotremes and marsupials.

Figure 34.42

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Exploring Mammalian Diversity
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Diagram for possible phylogenetic tree of mammals. All 20 extant orders of mammals are listed to right of tree.

A table is given:

Clade Orders	Description
Proboscidea, Sirenia, Tubulidentata, Hyracoidea, Afrosoricida, Macroscelidea	This clade of eutherians evolved in Africa when the continent was isolated from other landmasses. It includes Earth's largest living land animal (the African elephant), as well as species that weigh less than 10 grams.
Xenarthra	All members of this clade, which underwent an adaptive radiation in South America, belong to the order Xenarthra. One species, the nine-banded armadillo, is found in the southern United States.
Rodentia, Lagomorpha, Primates, Dermoptera, Scandentia	This is the largest eutherian clade. It includes the rodents, which make up the largest mammalian order by far, with about 1,770 species. Humans belong to the order Primates.
Carnivora, Cetartiodactyla, Perissodactyla, Chiroptera, Eulipotyphla, Pholidota	This diverse clade includes terrestrial and marine mammals as well as bats, the only flying mammals. A growing body of evidence, including Eocene fossils of whales with feet, supports putting whales in the same order (Cetartiodactyla) as pigs, cows, and hippos.

Another table is given for the evolutionary tree:

Orders and examples	Main characteristics
Monotremata: Platypuses, echidnas. Example shown is echidna.	Lay eggs; no nipples; young suck milk from fur of mother
Proboscidea: Elephants. Example shown is African elephant.	Long, muscular trunk; thick, loose skin; upper incisors elongated as tusks
Sirenia: Manatees, dugongs. Example shown is Manatee.	Aquatic; finlike forelimbs and no hind limbs; herbivorous
Xenarthra: Sloths, anteaters, armadillos. Example shown is tamandua.	Reduced teeth or no teeth; herbivorous (sloths) or carnivorous (anteaters, armadillos)
Lagomorpha: Rabbits, hares, picas. Example shown is Jackrabbit.	Chisel-like incisors; hind legs longer than forelegs and adapted for running and jumping; herbivorous
Carnivora: Dogs, wolves, bears, cats, weasels, otters, seals, walruses. Example shown is coyote.	Sharp, pointed canine teeth and molars for shearing; carnivorous
Cetartiodactyla, Artiodactyls: sheep, pigs, cattle, deer, giraffes. Example shown is bighorn sheep. Cetartiodactyla, Cetaceans: whales, dolphins, porpoises. Example shown is Pacific white-sided porpoise.	Artiodactyls: Hooves with an even number of toes on each foot; herbivorous. Cetaceans: Aquatic; streamlined body; paddle-like forelimbs and no hind limbs; thick layer of insulating blubber; carnivorous.
Marsupialia: Kangaroos, opossums, koalas. Example shown is koala.	Completes embryonic development in pouch on mother's body
Tubulidentata: Aardvarks	Teeth consisting of many thin tubes cemented together; eats ants and termites
Hyracoidea: Hyraxes. Example shown is rock hyrax.	Short legs; stumpy tail; herbivorous; complex, multi-chambered stomach
Rodentia: Squirrels, beavers, rats, porcupines, mice. Example shows red squirrel.	Chisel-like, continuously growing incisors worn down by gnawing; herbivorous
Primates: Lemurs, monkeys, chimpanzees, gorillas, humans. Example shows golden lion tamarin.	Opposable thumbs; forward-facing eyes; well-developed cerebral cortex; omnivorous
Perissodactyla: Horses, zebras, tapirs, rhinoceroses. Example shows Indian rhinoceros.	Hooves with an odd number of toes on each foot; herbivorous

Chiroptera: Bats. Example shows frog-eating bat.	Adapted for flight; broad skinfold that extends from elongated fingers to body and legs; carnivorous or herbivorous
Eulipotyphla: insectivores: some moles, some shrews. Example shows star-nosed mole.	Eat mainly insects and other small invertebrates

Evidence from numerous fossils and molecular analyses indicates that monotremes diverged from other mammals about 180 million years ago and that marsupials diverged from eutherians (placental mammals) about 140 million years ago. Molecular systematics has helped to clarify the evolutionary relationships between the eutherian orders, though there is still no broad consensus on a phylogenetic tree. One current hypothesis, represented by the tree shown below, clusters the eutherian orders into four main clades. The tree is represented by the following list.

Ancestral Mammal (Branch point 1)

Monotremes, 5 species; Order Monotremata (highlighted)

Branch point 2

Marsupials, 324 species; Order Marsupialia (highlighted)

Eutherians, 5010 species; Branch point 3

Orders Proboscidea (highlighted), Sirenia (highlighted), Tubulidentata (highlighted), Hyracoidea (highlighted), Afrosoricida (golden moles and tenrecs), Macroscelidea (elephant shrews)

Branch point 4

Order Xenarthra (highlighted)

Branch point 5

Orders Rodentia, Lagomorpha and Primates (all highlighted) and Dermoptera (flying lemurs) and Scandentia (tree shrews)

Orders Carnivora, Cetartiodactyla, Perissodactyla, Chiroptera and Eulipotyphla (all highlighted) and Pholidota (pangolins)

Primates

The mammalian order Primates includes the lemurs, tarsiers, monkeys, and apes. Humans are members of the ape group.

Derived Characters of Primates

Most primates have hands and feet adapted for grasping, and their digits have flat nails instead of the narrow claws of other mammals. There are other characteristic features of the hands and feet, too, such as skin ridges on the fingers (which account for human fingerprints). Relative to other mammals, primates have a large brain and short jaws, giving them a flat face. Their forward-looking eyes are close together on the front of the face. Primates also exhibit relatively well-developed parental care and complex social behavior.

The earliest known primates were tree-dwellers, and many of the characteristics of primates are adaptations to the demands of living in the trees. Grasping hands and feet allow primates to hang onto tree branches. All living primates except humans have a big toe that is widely separated from the other toes, enabling them to grasp branches with their feet. All primates also have a thumb that is relatively movable and separate from the fingers, but monkeys and apes have a fully **opposable thumb**; that is, they can touch the ventral surface (fingerprint side) of the tip of all four fingers with the ventral surface of the thumb of the same hand. In monkeys and apes other than humans, the opposable thumb functions in a grasping “power grip.” In humans, a distinctive bone structure at the base of the thumb allows it to be used for more precise manipulation. The unique dexterity of humans represents descent with modification from our tree-dwelling ancestors. Arboreal maneuvering also requires excellent eye-hand coordination. The overlapping visual fields of the two forward-facing eyes enhance depth perception, an obvious advantage when brachiating (traveling by swinging from branch to branch in trees).

Living Primates

There are three main groups of living primates: (1) the lemurs of Madagascar **(Figure 34.43)** and the lorises and bush babies of tropical Africa and southern Asia; (2) the tarsiers, which live in southeastern Asia; and (3) the **anthropoids**, which include monkeys and apes and are found worldwide. The first group—lemurs, lorises, and bush babies—probably resemble early arboreal primates. The oldest known tarsier fossils date to 55 million years ago; along with DNA evidence, these fossils indicate that tarsiers are more closely related to anthropoids than to the lemur group **(Figure 34.44)**.

Figure 34.43

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    Verreaux’s sifakas (*Propithecus verreauxi*), a type of
lemur.
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Lemur with long arms, legs and tail climbs a tree with a baby clinging to its back.

Phylogenetic tree for primates starting with ancestral primate and branching into the Anthropoids and the lemurs, lorises, bush babies and tarsiers. Timescale for the tree ranges from 70 million years ago to the present with increments every 10 million years. The tree is represented by the following list.

Ancestral Primate; Branch point 1; just under 60 million years ago

Lemurs, lorises and bush babies

Branch point 2; about 55 million years ago

Tarsiers

Anthropoids; Branch point 3; about 46 million years ago

New World monkeys

Branch point 4; about 28 million years ago

Old World monkeys

Branch point 5; about 16 million years ago

Gibbons

Branch point 6; about 13 million years ago

Orangutans

Branch point 7; about 8 million years ago

Gorillas

Branch point 8; about 6 million years ago

Chimpanzees and bonobos

Branch point 9; about 5 million years ago

Australopithecus

Humans and extinct Homo species

Figure 34.44

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A phylogenetic tree of primates.
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Phylogenetic tree for primates starting with ancestral primate and branching into the Anthropoids and the lemurs, lorises, bush babies and tarsiers with timescale.

The fossil record indicates that the lineage leading to monkeys and apes diverged from other primates 55 million years ago. The lineage leading to humans and *Australopithecus* branched off from other apes 7–8 million years ago; the dagger symbol † denotes an extinct group.

VISUAL SKILLS*Is the phylogeny shown here consistent with the idea that humans evolved from chimpanzees? Explain.*

*For suggested answer, see*Appendix A*.*

You can see in Figure 34.44 that monkeys do not form a clade but rather consist of two groups, the so-called "New" and "Old" World monkeys. Both of these groups are thought to have originated in Africa or Asia. The fossil record indicates that New World monkeys first colonized South America roughly 25 million years ago. By that time, South America and Africa had drifted apart, and monkeys may have reached South America from Africa by rafting on logs or other debris. What is certain is that New World monkeys and Old World monkeys underwent separate adaptive radiations during their many millions of years of separation **(Figure 34.45)**. All species of New World monkeys are arboreal, whereas Old World monkeys include ground-dwelling as well as arboreal species. Most monkeys in both groups are diurnal (active during the day) and usually live in bands held together by social behavior.

Figure 34.45

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New World monkeys and Old World monkeys.
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Part a. New World monkeys, such as spider monkeys (shown here), squirrel monkeys, and capuchins, have a prehensile tail (one adapted for grasping) and nostrils that open to the sides. Photo: Spider monkey with long limbs and tail perches in a tree.

Part b. Old World monkeys lack a prehensile tail, and their nostrils open downward. This group includes macaques (shown here), mandrills, baboons, and rhesus monkeys. Photo: Macaque with bushy, bright facial fur sits on a stump.

The other group of anthropoids consists of primates informally called apes **(Figure 34.46)**. The ape group includes four genera of gibbons, along with the so-called "great apes," the genera *Pongo* (orangutans), *Gorilla* (gorillas), *Pan* (chimpanzees and bonobos), and *Homo* (humans). The apes diverged from Old World monkeys about 25–30 million years ago. Today, nonhuman apes are found exclusively in tropical regions of the Eastern Hemisphere. With the exception of gibbons, living apes are larger than either New or Old World monkeys. All living apes have relatively long arms, short legs, and no tail. Although all nonhuman apes spend time in trees, only gibbons and orangutans are primarily arboreal. Social organization varies among

the apes; gorillas and chimpanzees are highly social. Finally, compared to other primates, apes have a larger brain in proportion to their body size, and their behavior is more flexible. These two characteristics are especially prominent in our next group, the hominins.

Figure 34.46

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Nonhuman apes.
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Nonhuman apes.

Parts are as follows.

Part a. Gibbons, such as this Muller's gibbon, are found only in southeastern Asia. Their very long arms and fingers are adaptations for brachiating (swinging by the arms from branch to branch).

Part b. Orangutans are shy apes that live in the rain forests of Sumatra and Borneo. They spend most of their time in trees; note the foot adapted for grasping and the opposable thumb.

Part c. Gorillas are the largest apes; some males are almost 2 m tall and weigh about 200 kg. Found only in Africa, these herbivores usually live in groups of up to about 20 individuals.

Part d. Chimpanzees live in tropical Africa. They feed and sleep in trees but also spend a great deal of time on the ground. Chimpanzees are intelligent, communicative, and social.

Part e. Bonobos are in the same genus (Pan) as chimpanzees but smaller. They survive today only in African nation of Congo. A bonobo stands on all fours with infant on her back.

Video

Chimp Cracking Nut

image

Video

Gibbons Brachiating